

“Energy efficiency is not just about consumption”



Germany-based Infineon Technologies AG offers semiconductor and system solutions addressing central challenges to modern society. The company's India office in Bengaluru is a major development centre of its R&D network, and plays a vital role in software development and hardware design.

Arunjai Mittal, Member of Management Board, Infineon Technologies AG, speaks to Jalaja Ramanunni and Dilin Anand of EFY on how energy efficiency can be best achieved and what Infineon Technologies is up to

ARUNJAI MITTAL
MEMBER OF MANAGEMENT BOARD, INFINEON TECHNOLOGIES AG

How important are energy efficiency, mobility and security in the future of technology?

All three are very important. Consider smart grids. A grid becomes smart only when it has the ability to choose sources of power, give consumers the option to utilise power at a certain tariff, etc. This is not possible without the use of intelligent powered semiconductors and secure solutions. Be it smart meters, AC-DC transmission lines, wind mills or solar applications, all of these need semiconductors.

How reactive is the Indian market to the need for energy efficiency?

It is pretty reactive and there is a high consciousness about it. The Bureau of Efficiency which has been set up is a confirmation that energy efficiency is an issue of importance to the policy makers.

In power distribution and transmission, people have been using AC transmission until now. Going forward, it is very clear that to be energy efficient, you need to use DC transmission. High-voltage DC is very popular. This trend is catching up in India too. In power generation, we have renewables and solar power is becoming very popular. On the consumption side, depending on the applications, there are many different markets.

What was the focus of your talk at the ISA Vision Summit?

I tried to bring forward to the industry that energy should be seen as a whole—from generation and distribution, to power consumption. All these have to be considered from an efficiency point of view. We cannot focus only on consumption and making our electronics more efficient. We have to consider transmission, distribution and generation as well. All three have to be considered if we are serious about energy efficiency.

Which of these consumes most energy in India?

Take a look at the chain of electricity—all the way from generation to distribution to consumption. You will see that all of them are generating losses. In power generation, a move from fossil fuel-based technology to renewables translates to a big gain in energy as energy losses are cut significantly. We can argue that this is the sector where energy losses can be minimised the most. On the other hand, if you move from AC transmission to DC transmission, you can reduce losses, indicating that transmission is also very important. Finally, on the consumption side, whether it is your charger or colour television, each of these plays a role in loss of energy as it sums up to a big number when added up. The amount of work that goes into making all these